

BMJ Open Comparative effectiveness of methadone take-home dose initiation in British Columbia, Canada: protocol for a population-based retrospective cohort study using target trial guidelines

Md. Belal Hossain ^{1,2}, Brenda Carolina Guerra-Alejos,¹ Megan Kurz,^{1,3} Jeong Eun Min,¹ Mohammad Ehsanul Karim,^{1,2} Shaun Seaman ⁴, Paxton Bach,^{5,6} Robert W Platt,⁷ Paul Gustafson ⁸, Julie Bruneau,^{9,10} Lawrence McCandless,^{3,11} Maria Eugenia Socías,^{5,6} Bohdan Nosyk ^{1,3}

To cite: Hossain MB, Guerra-Alejos BC, Kurz M, *et al*. Comparative effectiveness of methadone take-home dose initiation in British Columbia, Canada: protocol for a population-based retrospective cohort study using target trial guidelines. *BMJ Open* 2025;**15**:e095198. doi:10.1136/bmjopen-2024-095198

► Prepublication history and additional supplemental material for this paper are available online. To view these files, please visit the journal online (<https://doi.org/10.1136/bmjopen-2024-095198>).

Received 16 October 2024
Accepted 20 January 2025



© Author(s) (or their employer(s)) 2025. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group.

For numbered affiliations see end of article.

Correspondence to

Dr Bohdan Nosyk;
bnosyk@sfu.ca

ABSTRACT

Introduction Due to inferior safety profile and higher risk of diversion than buprenorphine/naloxone, guidelines typically recommend stringent eligibility criteria such as daily witnessed ingestion of methadone for at least 12 weeks before considering take-home doses. Recent research has focused on whether or not to initiate take-home methadone doses, often using pandemic-era data when temporary prescribing changes provided a natural experiment on the impact of access to take-home doses. However, none of these studies adequately examined the optimal timing and criteria for safely starting take-home doses to enhance treatment outcomes. To determine the optimal timing for initiating methadone take-home doses, we will compare the effects of different initiation times on time to treatment discontinuation, all-cause mortality and acute-care visits among individuals who completed methadone induction in British Columbia, Canada, from 2010 to 2022.

Methods and analysis We propose emulating a target trial using linked population-level health administrative data for all individuals aged 18 or older living in British Columbia, Canada, completing methadone induction between 1 January 2010 and 31 December 2022. The exposure strategies will include no take-home dosing and take-home dose initiation in ≤4, 5–12, 13–24 and 25–52 weeks since completed induction. The outcomes will include the time to treatment discontinuation, all-cause mortality and acute-care visits. We propose a per-protocol analysis with a clone-censor-weighting approach to address the immortal time bias implicit in the comparison of alternative take-home dose initiation times. Subgroup and sensitivity analyses, including cohort restrictions, study timeline variations, eligibility criteria modifications and outcome reclassifications, are proposed to assess the robustness of our results.

Ethics and dissemination The protocol, cohort creation and analysis plan have been classified and approved as a quality improvement initiative by Providence Health Care Research Ethics Board and the Simon Fraser University Office of Research Ethics. Results will be disseminated to

STRENGTHS AND LIMITATIONS OF THIS STUDY

- ⇒ A target trial using observational data will be emulated to determine the effectiveness of different initiation times of methadone take-home doses as observed in clinical practice in British Columbia.
- ⇒ The per-protocol analysis with a clone-censor-weighting approach will be used to address immortal time bias and confounding by indication.
- ⇒ Potential confounding bias and other threats to validity will be assessed via a range of sensitivity and subgroup analyses.
- ⇒ One limitation of this study is that health administrative databases do not provide information on whether patients are diverting their methadone medication to non-prescribed uses.
- ⇒ Residual confounding remains possible due to the observational nature of the data, while sensitivity analyses with and without instrumental variable approach will be conducted to address residual confounding bias.

local advocacy groups and decision-makers, national and international clinical guideline developers, presented at international conferences and published in peer-reviewed journals.

INTRODUCTION

Opioid agonist treatment

Following the introduction of fentanyl into the illicit drug supply,^{1 2} North America has experienced a rapid increase in opioid-related deaths—from 43 149 in 2016 to 81 083 in 2023 in the USA and from 2831 to 5975 in the same time period in Canada.^{3–6} Opioid agonist treatment (OAT), such as methadone and buprenorphine/naloxone, is an evidence-based treatment for people with opioid use disorder (OUD). Both medications have been

shown to be effective in reducing the harms associated with OUD, including reducing the risk of hospitalisation and death.^{7–11} However, retention in treatment in British Columbia (BC), Canada, and elsewhere internationally is a persistent challenge. Annual retention rates in BC have been declining since 2014,¹² coinciding with the higher-potency opioids in the illicit drug market.¹³ Nevertheless, all aspects of OAT prescribing and the barriers to access and achieving successful stabilisation of these WHO essential medicines require re-evaluation to ensure their benefits are being fully realised given the influx and epidemic of fentanyl and analogues.

Supervised and take-home dosing

A major barrier to OAT retention is the requirement for prolonged daily supervised dosing.¹⁴ Studies reported that allowing clients to receive take-home doses could improve OAT adherence and retention in treatment.^{15 16} However, international clinical guidelines, including Canadian guidelines, recommend daily witnessed ingestion of methadone under the supervision of a healthcare professional until clients demonstrate clinical stability.^{17–20} According to the 2017 BC OUD guideline, methadone clients in BC were considered eligible for take-home doses if they demonstrated social stability and long-term abstinence from illicit drug use with a minimum of 4 weeks of stable dosing and a minimum of twelve weeks of urine drug testing in addition to the clinical stability.²¹ Homelessness, inability to safely store medications, ongoing substance use (eg, benzodiazepines, alcohol, other sedatives), concurrent mental health conditions, and history of diversion were all noted as factors discouraging take-home dosing.²¹ Updated guidelines, released in 2023, made few changes such as clients can receive take-home doses as soon as clinical stability is demonstrated for 4 weeks instead of 12.¹⁹ The stability is now also judged by the clinician but not based on strict criteria, and the urine drug testing requirement is also relaxed. However, clients have reported that supervised ingestion increases stigma and negatively impacts overall treatment engagement.¹⁴ Moreover, the requirement of daily attendance at a pharmacy, additional fees not covered through health insurance²² and potential travel requirements pose major challenges to clients. Although BC covers OAT and additional fees (eg, pharmacy dispensation and supervised ingestion),^{23–25} the excess cost of supervised ingestion was estimated to exceed \$40 million CAD annually for the government, with over 30% allocated towards fees for daily witnessed ingestion.²⁶

For methadone, take-home dosing requires a persistently high degree of clinical stability, requiring weeks or months to achieve.^{27 28} Although standards differ between provinces in Canada, meeting stringent eligibility criteria is necessary to receive methadone take-home doses.^{18 21 29–48} In BC, the guideline for take-home methadone doses starts with one dose per week, with the first dose administered under daily witnessed ingestion.^{19 21} Clients gradually progress to additional weekly take-home

doses based on clinical stability. In the USA, methadone programmes have rigorous guidelines and policies such as no evidence of illicit drug use, being mentally stable and having a safe place to store the treatment, similar to Canadian guidelines.^{17 49–51} The recommendations for take-home doses are approximately similar in other countries, such as Australia,^{52–54} New Zealand⁵⁵ and the UK.^{56 57} However, the guidance and supporting evidence on the optimal timing for initiating take-home dosing in provincial, national or international clinical guidelines is largely insufficient (table 1). As there is no set timeline for when clients are eligible to receive their first take-home dose after completed induction, clinically and socially stable clients could receive their first methadone take-home dose after 2–3 months of supervised ingestion.^{21 58} Ultimately, take-home methadone doses are available per the treating prescriber's judgement.⁵⁹

Compared with methadone, clients can receive buprenorphine/naloxone take-home doses as early as their first day of treatment, given that clinical stability may be achieved as early as 1–3 days.^{21 60} Due to a faster stabilisation period and the anti-abuse and diversion component of naloxone,⁶¹ flexible take-home buprenorphine/naloxone dosing schedules earlier in the course of treatment are also recommended in most settings including BC.^{18 21 39 62 63} However, the optimal time to initiate methadone take-home doses to promote OAT retention, conditioning on eligibility criteria, is unknown. We found no studies, experimental or non-experimental, that explicitly evaluated this clinical decision.

Impact of COVID-19 on take-home dosing policies

The COVID-19 pandemic prompted providers and policymakers to take steps to expand client eligibility for take-home dosing in order to help promote self-isolation. Countries including the USA,⁴⁹ Canada,⁶⁴ Australia,⁶⁵ England,⁶⁶ Italy⁶⁷ and Spain⁶⁸ relaxed guidelines to encourage the use of take-home doses. These included allowing take-home dosing regardless of the clinical stability or time in treatment of the clients.^{17 49 50 64–67} For example, the Substance Abuse and Mental Health Services Administration (SAMHSA) implemented a provision permitting prescribers to provide a maximum of 28 consecutive days of take-home methadone doses to stable patients and up to 14 days for those deemed less stable.⁴⁹ The new SAMHSA guideline published in 2024 also allows individuals to receive up to 7 days of take-home doses on the day treatment is initiated.⁶⁹ The only exception is that individuals who recently encountered an overdose, exhibited unstable psychiatric comorbidity or engaged in high-risk patterns of illicit substance use were deemed ineligible for take-home doses in at least Ontario.⁶⁴

Available evidence and gap in the literature

The evidence in the literature on the most effective and safe timing of take-home methadone dosing is scarce, that is, how clinical stability should be defined and at

Table 1 Guideline comparison of initiating take-home doses for methadone.

Guideline	Recommendation on the timing of take-home dosing	Supporting references
BC, Canada (2017) ²¹	Minimum 4 weeks of stable dosing and twelve weeks negative urine drug testing (UDT).	None
Alberta, Canada (2022) ³⁰	Minimum 3 months on methadone and three consecutive negative UDT.	None
Saskatchewan, Canada (2020) ³¹	Negative UDT for 3 months.	None
Manitoba, Canada (2024) ^{32 33}	Minimum 2 months on methadone unless criteria are met for Sunday take-home doses due to pharmacy closures.	None
Ontario, Canada (2011) ³⁴	Minimum 2 months on methadone, and at least 1 week without problematic substance use.	None
Ontario, Canada (2023) ²⁹	At least 4 weeks of treatment, clinically and psychologically stable, safe living environment and no overdose in the last month.	Sordo et al. (2017), ⁷ Zlotnick et al. (2003), ⁴⁶ Bouhnik et al. (2002), ⁴⁷ Akreét al. (2021) ⁴⁸
Prince Edward Island, Canada (2012) ³⁵	Minimum 2 months on methadone, and at least 1 week without problematic substance use	Ontario (2011) ³⁴
Quebec, Canada (2018), ¹⁸ New Brunswick, Canada (2018) ³⁶	Minimum 3 months on methadone.	CRISM (2018), ¹⁸ BC guideline (2017) ²¹
Nova Scotia, Canada (2012), ³⁷ Newfoundland and Labrador, Canada (2022) ³⁸	Minimum 3 months on methadone, four consecutive weeks of negative UDT and 2 months without substance use.	None
CRISM, Canada (2018) ¹⁸	Province-specific recommendation listed above.	Provincial guidelines ^{21 30–38}
SAMHSA, USA (2020), ⁴⁹ (2021) ³⁹	No specific timing but those who demonstrate clinical stability could receive the take-home doses.	None
SAMHSA, USA (2024) ⁶⁹	Clients can receive take-home doses on the treatment initiation day.	None
ASAM, USA (2020) ⁴¹	No specific timing but those who demonstrate treatment progress and are judged to be at low risk for diversion could receive the take-home doses.	None
CSAM, California, USA (2019) ⁵¹	Minimum 3 months on methadone unless criteria are met for Sunday and holiday take-home doses.	None
Australia (2014) ⁵²	No specific timing but those who demonstrate clinical stability could receive the take-home doses.	None
Western Australia, Australia (2014) ⁵³	Continuous treatment for a minimum of 6 months.	None
New South Wales, Australia (2018) ⁵⁴	Continuous treatment for 3 months.	Australia national guideline (2014) ⁵²
New Zealand (2014) ⁵⁵	No specific timing but those who demonstrate clinical stability could receive the take-home doses.	None
UK (2017) ⁵⁶	Clients could receive the take-home doses if they demonstrate clinical stability unless criteria are met for Sunday and holiday take-home doses.	None

Continued

Table 1 Continued

Guideline	Recommendation on the timing of take-home dosing	Supporting references
NICE, UK (2022) ⁵⁷	After 3–6 months on treatment and demonstrate clinical stability.	None

ASAM, American Society of Addiction Medicine; BC, British Columbia; CAMH, Centre for Addiction and Mental Health; CRISM, Canadian Research Initiative on Substance Misuse; CSAM, California Society of Addiction Medicine; NICE, National Institute for Health and Care Excellence; SAMHSA, Substance Abuse and Mental Health Services Administration; UDT, urine drug test.

what duration stability should be demonstrated before allowing take-home doses. Previous studies focused on whether or not take-home doses should be initiated, while none specifically addressed the optimal time to start. Studies mostly used COVID-era data to show the comparative effectiveness of receiving take-home methadone doses versus not receiving take-home doses. Using data from Connecticut, USA, Brothers *et al*⁷⁰ demonstrated no increase in the risk of overdose due to receipt of take-home doses during the COVID-19 pandemic. Studies also showed that relaxing restrictions on methadone take-home dosing during the pandemic improved client experience and OAT retention,¹⁵ client satisfaction, treatment access and engagement in care.^{71 72} However, the evidence is mixed as a cross-sectional study from the USA reported no association of methadone take-home dosing on treatment discontinuation and emergency department visits.⁷³ Further, a randomised clinical trial from Scotland reported no differences in treatment retention and adherence between supervised and take-home methadone doses.⁷⁴ The study also found no clinically significant increase in the 6-month mortality for flexible take-home dosing. Nonetheless, there is sparse evidence on the appropriate timing for receiving methadone take-home doses on treatment discontinuation and mortality. While the present study focused on the timing of the first take-home initiation, our group is subsequently determining the comparative effectiveness of different take-home dosing frequencies, such as taking home doses for different numbers of days, as observed in clinical practice in BC.

Clients’ perspectives on take-home doses

In practice, the assessment of clinical stability of a client is a continuous process which requires consideration of individual circumstances before allowing take-home doses. Factors such as clients’ clinical course, disease severity, treatment adherence and other social factors such as family support, financial status and housing status might influence the allowance of take-home doses.^{16 19 21 75 76} Receiving take-home doses can also be influenced by the availability of pharmacies, including those with limited opening hours or Sunday closures. In contrast, OAT prescribers hold mixed attitudes regarding regulations around take-home dosing of methadone.⁷⁷ Although client-centred care emphasises the need to focus on individual-level outcomes such as patient safety

and client treatment goals,^{78 79} studies from the USA showed that while daily witnessed ingestion was perceived as an unnecessary burden by some providers, many were reluctant to deviate from the strict guidelines.^{78 79} Clients have reported that regulations lead to views of paternalistic care over the clients’ choices in addition to feelings of stigma and moralism.^{78 80} Although public safety and diversion are large concerns in clinical management, initiation and eligibility of receiving take-home doses,^{19 21} poor adherence and declining retention rates in OAT should warrant assessing individual-level effects of treatment outcomes. While the majority of studies conducted during COVID-19 suggested equal or improved individual-level outcomes with increased access to take-home doses,^{15 70–74} two studies reported an increased rate of methadone-involved overdose deaths.^{42 81}

Study objectives

Our aim is to determine the comparative effectiveness of different initiation times of methadone take-home doses (within ≤4, 5–12, 13–24 and 25–52 weeks after methadone induction compared to no take-home doses) on the time to methadone discontinuation, all-cause mortality and acute-care visits as observed in clinical practice. Given that take-home dosing is designed to ease a noted barrier to prolonged retention in treatment, we hypothesised that individuals receiving earlier take-home doses would have lower rates of discontinuation, mortality and acute-care visits compared with those who did not. We will emulate a hypothetical target trial using data from population-based linked health administrative databases from BC, Canada (2010–2022).

METHODS

Study setting and data sources

We will execute a population-based retrospective cohort study in BC, Canada, between 1 January 2010 and 31 December 2022. The cohort will be constructed based on a linkage of nine health administrative databases, including Client Roster (capturing demographic and geographic information),⁸² PharmaNet (capturing drug dispensations),⁸³ Medical Services Plan (capturing physician billing records),⁸⁴ Discharge Abstract Database (hospitalisations, inpatients and day surgeries),⁸⁵ National Ambulatory Care Reporting System database (capturing all emergency department visits),⁸⁶ BC

Provincial Corrections (capturing incarceration in provincial prisons),⁸⁷ Perinatal Database (capturing maternal and child health for all provincial births),⁸⁸ Social Development and Poverty Reduction Database (capturing social assistance receipt)⁸⁹ and Vital Statistics (capturing all deaths in the province).⁹⁰ The databases are linked using a de-identified individual-level unique identifier,⁹¹ as done in previous studies.^{92–94} Information on some time-dependent covariates, such as unstable housing and income assistance, was unavailable before 1 January 2010. Additionally, at the time of protocol submission, our data cut-off date was 31 December 2022. Therefore, we plan to restrict our analyses to the period from January 2010 to 31 December 2022.

Study population

The study will include individuals aged 18 years and older who completed a methadone induction, where we will define ‘completed induction’ as the date of reaching the end of 2 weeks of continuous treatment with no dose changes as a measure of clinical stability.⁹⁵ Those who received take-home doses before time zero (completed methadone induction) will be excluded. Individuals with a prescription of other OAT, such as prescription of buprenorphine/naloxone and slow-release oral morphine, prior to completed methadone induction will also be excluded. Pregnant women, those who are incarcerated and individuals receiving cancer treatment or palliative care at time zero will also be excluded from the analysis. We will exclude pregnant women since the pharmacokinetics of methadone are altered during pregnancy, and pregnant women often need expert guidance on treatment management.^{96 97} Those who complete induction while incarcerated will be excluded since they are not eligible to receive a take-home dose during incarceration.¹⁹ As methadone has been used in pain management for cancer⁹⁸ and in palliative care,⁹⁹ eligibility for take-home doses among this population is different from individuals with OUD. If individuals become pregnant, incarcerated, receive cancer treatment or palliative care, or switch to a different OAT after time zero but during follow-ups, they will be censored.

Study design

We will emulate a target trial using our observational data. [Table 2](#) summarises the key components of the target trial emulation. We will execute both incident user and prevalent new-user study designs to ensure the clinical experience of those accessing treatment in successive attempts are captured in our analyses ([figure 1](#)). Incident users include individuals engaged in methadone for the first time during the study period without any prior methadone experience dating back to 1 January 1996. To define prevalent new users, we will use a 30-day washout period, entailing no OAT dispensations in the 30 days prior to the episode initiation.⁹³

Study follow-up

The date of completed methadone induction will be the time zero of the study. The study will be executed using

a weekly counting process notation, following individuals from time zero until the primary or secondary outcomes (defined below), censoring due to protocol violations (ie, becoming pregnant, incarcerated, starting cancer treatment or palliative care, or switching to a different OAT), taper initiation (defined as the date of second weekly dosage decrease with no interim increases), out-migration from the province or end of follow-up (31 December 2022). Out-migration will be defined as de-registration from the Medical Services Plan and an indication of being out of the province at the client’s residence.

Exposure

The exposure of interest is the time from completed induction to initiation of methadone take-home doses. The exposure strategies will include: (i) no take-home dose and take-home dose in (ii) ≤ 4 weeks, (iii) 5–12 weeks, (iv) 13–24 weeks and (v) 25–52 weeks since time zero (completed induction). These early initiations of take-home dosing strategies, for example, within 12 weeks, are defined to accommodate the BC guideline recommendations for take-home dose receipt after 2–3 months of supervised ingestion in the 2017 guideline^{21 58} and 4 weeks in the 2023 guideline;¹⁹ alternative strategies—both sooner and later than recommended—are defined to serve as contrasts to current recommendations. In other words, these additional cut points of 13–24 weeks and 25–52 weeks are chosen to comprehensively provide a thorough analysis of different initiation times. Compared with the guidelines, which recommended receiving take-home doses after 4 weeks¹⁹ or 2–3 months since OAT initiation,^{21 58} our exposure strategies are defined since the completed induction. Considering ‘completed induction’ as time zero allows all participants to be eligible to receive take-home doses. We will define take-home dosing initiation as the start of take-home methadone dosing for ≥ 2 days regardless of dosage after completed induction. The 2-day threshold is chosen to account for weekend take-home dosing in rural regions with Sunday pharmacy closures.

Outcomes

The primary outcome of interest is the time to methadone discontinuation, defined as a break in prescribed doses lasting more than 5 days. This discontinuation definition is based on the BC guideline recommending reassessment and restarting titration after five consecutive missed doses.²¹ As our data do not identify OAT receipt in inpatient settings, we assumed individuals who began OAT prior to admission continued treatment during their hospitalisation. The secondary outcome will be the time to all-cause mortality, where the follow-up will begin at time zero and end at death or censoring defined above. We will further consider drug-related acute-care visits and overdose-related acute-care visits as secondary outcomes. Acute-care visits will be defined through hospitalisations or emergency department visits. The estimand is defined in [table 3](#).

Table 2 Key components of the protocol of the target trial of the time to take-home dose initiation.

Component	Target trial protocol	Emulated trial using observational data
Eligibility criteria	Individuals aged ≥ 18 years with opioid use disorder who completed methadone induction with no take-home, with the following eligibility: non-pregnant or not planning for pregnancy, not incarcerated and no history of receiving cancer treatment or palliative care.	Incident user design: Individuals aged ≥ 18 years with opioid use disorder who completed methadone induction and no prior experience of receiving opioid agonist treatment (OAT) dating back to 1 January 1996, with the following eligibility: non-pregnant, not incarcerated and no history of receiving cancer treatment or palliative care. Prevalent new user design: selection criteria above, with no history of receiving OAT within the past 30 days.
Exposure strategies	Time from completed induction such as stable dosing and elimination of withdrawal symptoms to take-home dose initiation in different time intervals or no take-home dose (control).	Time from completed induction such as reaching the end of a 2-week period with no dose increase to take-home dose initiation in ≤ 4 weeks, 5–12 weeks, 13–24 weeks and 25–52 weeks, or no take-home dose (control).
Exposure assignment	Individuals are randomly assigned to the exposure strategies at completed induction and are aware of the strategy to which they are assigned.	Exposure strategies defined above are assigned (via clones) to all the strategies compatible with their data at completed induction.
Outcomes	Methadone discontinuation (primary), all-cause mortality, drug-related acute-care visits and overdose-related acute-care visits (secondary).	Same. Treatment discontinuation will be defined as a break in prescribed doses lasting more than 5 days. Mortality information will be observed via linkage to vital statistics records. Acute-care visits will be defined through hospitalisations or emergency department visits.
Follow-up	Follow-up begins at completed induction and ends in treatment discontinuation, death, study end, violation of eligibility criteria after time zero or intercurrent events.	Same.
Causal contrast	Per-protocol.	Same.
Analysis plan	Individuals are censored when they deviate from their assigned exposure strategy, and potential selection bias is adjusted for using inverse probability weighting.	Same per-protocol analysis, except that individual clones are used.

Analysis plan

Primary analysis

Since initiation of take-home dosing is not observed at time zero, incorrectly defining this time-dependent exposure variable as a time-fixed/baseline variable leads to immortal time bias.^{100–102} It has been shown that immortal time bias produces a biased effect estimate.^{102 103} We will conduct a per-protocol analysis with the clone-censor-weighting approach¹⁰⁴ to address the risk of immortal time bias in evaluating the effectiveness of alternative times of initiating take-home doses on methadone discontinuation, mortality and acute-care visits. We will create five clones of each individual, with one clone for each exposure strategy. By definition, the study arms are identical with respect to baseline characteristics and compatible with their data at cohort entry. Hence, the bias due

to baseline confounding will be removed.^{104 105} In each exposure strategy, we will censor individuals when they deviate from their assigned exposure strategy (figure 2). Although cloning will allow us to account for baseline confounding by indication, artificial censoring introduces time-dependent confounding. To address the bias introduced by informative censoring, we will use inverse-probability-of-censoring weighting (IPCW).^{104–106} Pooled logistic regression will be used to predict the individual probabilities of remaining uncensored at each week.¹⁰⁷ The model will include the list of measured time-fixed and time-dependent covariates described above. Stabilised weights will be calculated to prevent extreme weights¹⁰⁸:

$$SW_{it} = \prod_{j=1}^t \frac{\Pr(C_{ij}=01B_i, \bar{C}_{ij-1}=0)}{\Pr(C_{ij}=01B_i, L_{ij}, C_{ij-1}=0)},$$

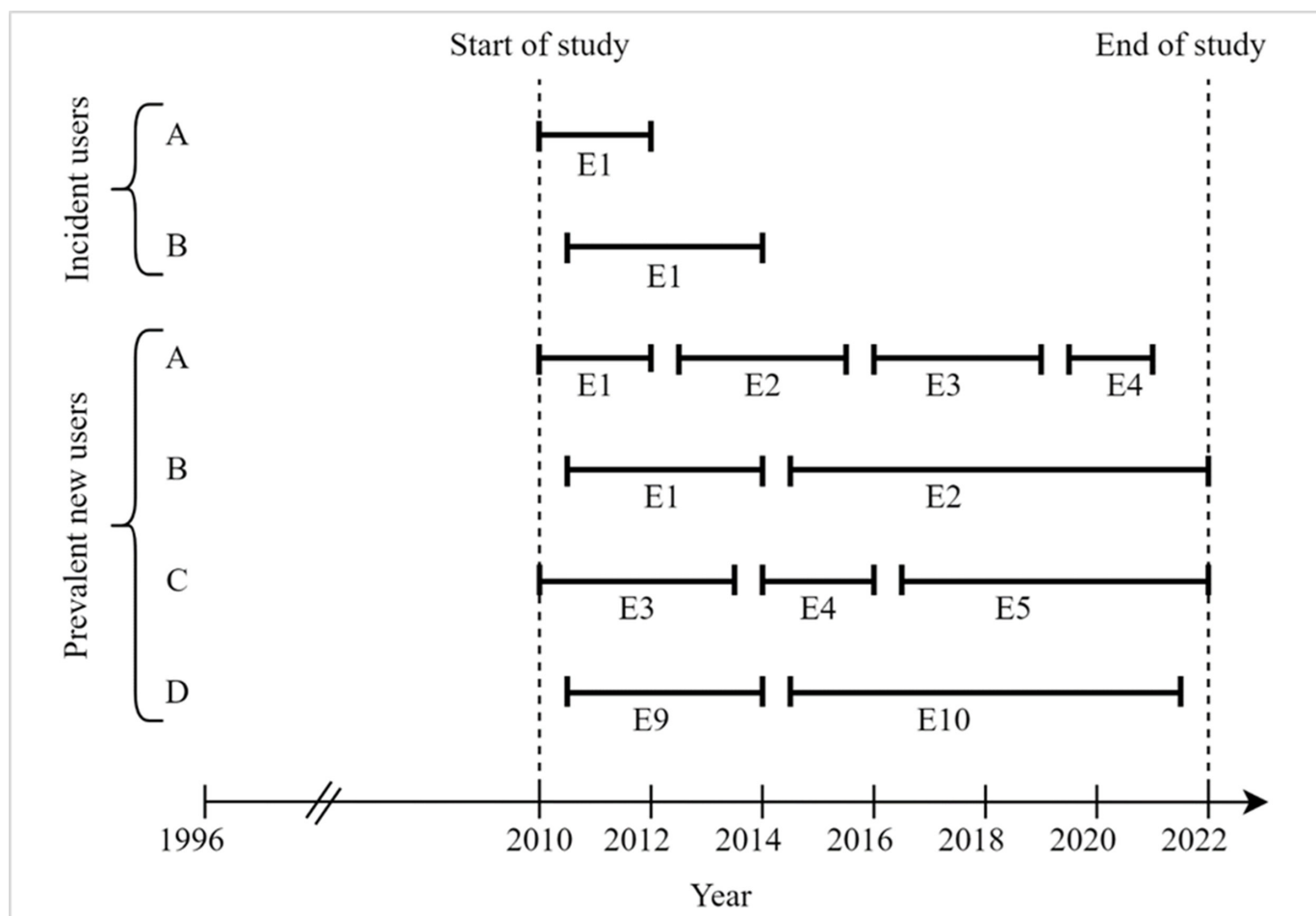


Figure 1 An example of incident users and prevalent new users for four individuals (A–D) in exploring the relationship between times for take-home methadone dosing and methadone discontinuation. Here, E_k indicates the k^{th} episode number. Legend: The first episodes of individuals A and B are part of the incident user design, while all episodes for these two individuals during the study period are part of the prevalent new-user design. There were two episodes before 2010 for individual C and eight episodes for individual D. The prevalent new-user design includes episodes 3–5 for individual C and episodes 1–2 for individual D. Episodes 1–2 for individual C and episodes 1–8 for individual D are not considered since these episodes are prior to the study start date. An episode is defined as the start of methadone initiation to methadone discontinuation or censoring. To define prevalent new users, a 30-day washout period, such as no opioid agonist treatment dispensations in the 30 days prior to the episode initiation, will be used.

where SW_{it} is the stabilised weight for the i th individual at time t , C_{ij} is the artificial censoring status for the i th individual at j th time, $\bar{C}_{ij}$ is the artificial censoring history for the i th individual at j th time, B is the list of time-fixed covariates and $\bar{L}_{ij}$ is the time-dependent covariate history for the i th individual at j th time. Time-dependent covariate (defined below) will be lagged by a week such that censoring will be predicted by the time-varying covariates from the prior week. Notably, this analysis assumes that regular data were collected for the time-dependent covariates. However, we will use linked health administrative databases to define the covariates. Thus, the time-dependent covariates will be measured based on the presence or absence of codes used in determining those covariates in a specific week, which is analogous to the last observation carried forward approach.^{109 110} We will examine the distribution of the weight to see whether the mean is approximately 1. If we observe extreme weights,

we will truncate them at a smaller level (eg, 99th percentile) to avoid undue influence of outliers.¹¹¹ In cases where we truncate the weights, we will report both the untruncated and truncated weights.

A second set of weights will be calculated to account for censoring due to protocol violations such as becoming pregnant, incarcerated, starting cancer treatment or palliative care, taper initiation or out-migration from the province. A third set of weights will be calculated to account for treatment switching. We will use the same set of covariates to calculate the stabilised weights of censoring on the original dataset before cloning. As described by Maringe *et al*,¹⁰⁵ three sets of weights will be multiplied. In other words, the final weight for each individual time t will be the product of the ‘clone censoring weight’ and ‘censoring weight due to protocol violations’ and ‘censoring weight due to treatment switching’ for an

Table 3 Specification of the estimand in exploring the relationship of times for take-home methadone dosing with methadone discontinuation, mortality and acute-care visits.

Component	Primary outcome: methadone discontinuation	Secondary outcomes: mortality and acute-care visits
Population	Individuals aged ≥ 18 years with opioid use disorder who completed methadone induction, such as reaching the end of a 2-week period with no dose increase, with no prior take-home methadone doses, non-pregnant or not planning for pregnancy, not incarcerated and not receiving cancer treatment or palliative care.	The same.
Treatment conditions	Control group: no take-home dose Comparator groups: take-home dose in ≤ 4 weeks, (iii) 5–12 weeks, (iv) 13–24 weeks and (v) 25–52 weeks since completed induction.	The same.
Endpoint	The first occurrence of a discontinuation, such as a break in prescribed methadone doses lasting more than 5 days.	(i) all-cause mortality, (ii) drug-related acute-care visits, (iii) overdose-related acute-care visits.
Summary measure	HR.	The same.
Handling of intercurrent events	Censor at intercurrent events (taper initiation, switch to another opioid agonist treatment)	The same.
HR, Hazard ratio.		

individual up to that time point. After calculating the weights, we will fit the pooled logistic regression with the final weights to explore the relationships of time to initiating take-home doses with methadone discontinuation, mortality and acute-care visits. Since we will use the stabilised weights to explore the relationships, we will adjust the outcome models for the time-fixed covariates.¹¹² White's robust sandwich estimator will be used to calculate the SE and the 95% compatibility/confidence interval (95% CI).¹¹³ The HR with 95% CI will be reported. To look at the methadone discontinuation, mortality and acute-care visit rate over follow-ups, we will present the cumulative incidence curves, with 200 bootstrap samples to obtain the 95% CI.^{105 114} We will report the risk difference with a 95% CI at the 1, 2, 3, 4 and 5-year follow-ups.

While we will use IPCW to deal with informative censoring, the following conditions must be met to obtain unbiased estimates of time to initiating take-home doses on OAT discontinuation, all-cause mortality and acute-care visits: (i) no unmeasured/uncontrolled confounding, (ii) correct model specification of the censoring model, (iii) positivity and (iv) causal consistency.^{104–106} We will evaluate and address these conditions as part of the analysis to minimise bias and ensure the robustness of the results. First, unmeasured covariates could be variables other than the time-fixed and time-dependent covariates (described below) that might influence prescribers' decisions to approve take-home doses. While the

assumption of no uncontrolled confounding cannot be verified in observational studies, we will adjust for all potential confounders available within our linked health administrative databases (table 4). Second, we will explore whether the stabilised weights have a mean of one, which is necessary for correct model specification.¹¹⁵ Third, we will empirically verify the artificial censoring status in terms of their covariate values, such as assessing whether there are any subgroups of the population with zero proportion of being uncensored, as this could indicate violations of the positivity assumption.^{112 116} Although IPCW are known to be more stable than inverse probability of treatment weighting,¹¹⁷ we will apply techniques such as truncation of the censoring weights at predefined percentiles (99th) in cases where violations of the positivity assumption are detected. Fourth, the causal consistency assumption could be violated since we define our exposure variable in terms of intervals. For example, grouping individuals for take-home dosing at 13 weeks and 24 weeks together could introduce heterogeneity and differential risk profiles that may violate the causal consistency assumption. While the specific cut points of 13–24 and 25–52 weeks may not be based on strong clinical guidelines, individuals who initiate take-home doses later in the treatment process tend to share similar characteristics. Therefore, the potential heterogeneity within these groupings is expected to be minimal and unlikely to significantly impact the validity of our findings.

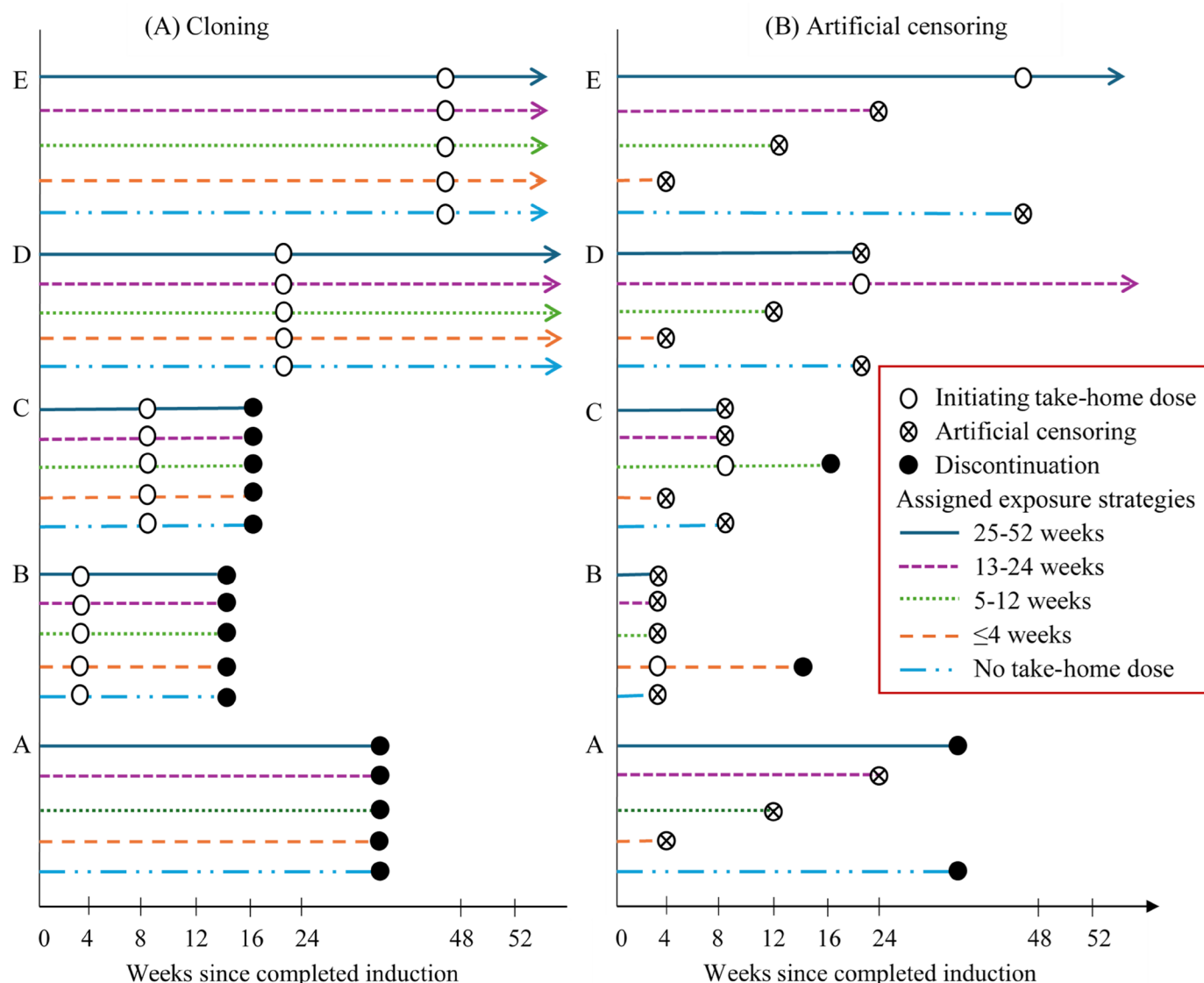


Figure 2 Schematic illustration of clone censoring approach for five individuals (A–E) in exploring the relationship between times for take-home methadone dosing and methadone discontinuation.

Legend: **(A)** five clones (one for each exposure strategy) for five individuals, where exposure strategies are defined as the time from completed induction (time zero) to methadone take-home dose. **(B)** Clone censoring, where individuals are censored for other exposure strategies at the end of each time interval and censored at the same time, with actual take-home dose initiation for the strategy at a longer time interval. For example, for the ‘no take-home dose’ arm, individuals B–E are censored at the time they receive take-home doses. Similarly, for the ‘5–12 weeks’ arm, individuals A, B, D and E are censored when they receive a take-home dose before 5 weeks (B), after 12 weeks (D, E) or no take-home dose (A).

To allow less variability in the treatment effect, we choose 13–24 and 25–52 weeks instead of grouping 13–52 weeks.

Covariates

We will identify the confounders based on a review of the literature^{16 75 76 92 93 118} and a causal diagram (figure 3). Time-fixed and time-dependent covariates defined in table 4 and unmeasured covariates could influence prescribers’ decisions to approve take-home doses. However, we can measure and sufficiently control for the proposed time-fixed/baseline and time-dependent covariates that influence the relationship between initiation of take-home doses and the outcomes. Time-fixed

covariates will be defined at time zero (completed induction). Time-dependent covariates will be defined in a 1 year look-back period and updated weekly.

We previously conducted a systematic review of published articles to identify the factors associated with OAT discontinuation.⁹² We propose to augment this list of time-fixed and time-dependent covariates with additional variables available in our linked administrative databases that are hypothesised to influence the initiation of take-home dosing, methadone discontinuation, mortality and acute-care visits. The list of time-fixed covariates includes the demographic variables, while time-dependent covariates include measures of socioeconomic status, medical

Table 4 Potential confounding variables influencing the relationship of times for take-home methadone dosing with treatment discontinuation, all-cause mortality and acute-care visits.

Covariate	Definition
Demographic (time-fixed*)	
Age	Age in years (continuous) at completed induction.
Sex	Whether male or female.
Place of residence	Rural or urban place of residence at completed induction.
Calendar year (time-fixed*)	
Year of completed induction	2010–2012, 2013–2015, 2016–2019, 2020–2022
Socioeconomic (time-dependent†)	
Unstable housing	Whether homeless, inadequately housed or not, which will be constructed using data from Medical Services Plan, Discharge Abstract Database, and Development and Poverty Reduction Database.
Receipt of income assistance	Whether received any income assistance.
Medical conditions (time-dependent†)	
Charlson Comorbidity Index	Continuous, which is developed based on 17 concurrent conditions from hospital separations databases: myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, rheumatic disease, peptic ulcer disease, mild liver disease, moderate/severe liver disease, diabetes without chronic complication, diabetes with chronic complication, hemiplegia or paraplegia, renal disease, malignancy, metastatic solid tumour and HIV/AIDS.
Chronic pain	Whether had chronic pain.
Behavioural and care-related (time-dependent†)	
Non-opioid substance use disorder	Whether had any substance use disorder other than opioid use disorder.
Alcohol use disorder	Indication of alcohol use disorder.
Tobacco use disorder	Indication of tobacco use disorder.
Severe mental disorder	Indication of severe mental disorder such as major depressive disorder, bipolar disorder, or schizophrenia.
Sedative medication	Indicator for receipt of sedative medication.
Prescription opioid other than OAT	Indicator for receipt of prescription opioid other than OAT.
Psychiatric medication	Indicator for receipt of psychiatric medication.
Mental healthcare	Indicator of mental healthcare excluding severe mental disorder.
Care from community health centres (time-fixed*)	Whether received care from community health centres.
Hospitalisation (time-dependent†)	
Drug-related hospitalisations	Indicator for drug-related hospitalisations or emergency department visits (except for acute-care visit outcomes).
Psychiatric hospitalisations	Indicator for psychiatric hospitalisations.
OAT-related (time-dependent†)	
Time to completed induction	Time from methadone initiation to completed induction in days (continuous).
Methadone interruption	Indicator for interruption in methadone in the previous week.
Methadone dose level	Methadone dose level in the previous week (<80 mg, 80–120 mg, >120 mg)
Mode of delivery (time-dependent†)	
Use of virtual care	Whether used virtual care
Receipt of OAT by delivery	Whether pharmacy delivered

Continued

Table 4 Continued

Covariate	Definition
Short-acting opioids alongside methadone	Whether concurrent receipt of short-acting opioids such as hydromorphone tablets alongside methadone.
*Time-fixed covariates will be defined at the time of completed induction.	
†Time-dependent covariates will be updated weekly; indication of chronic medical conditions will be fixed after initial identification.	
OAT, opioid agonist treatment.	

conditions, behavioural and care-related, hospitalisation and methadone-related variables (table 4). Measurable indicators of clinical instability include drug-related acute-care visits, indication of homelessness and potentially interruptions (<5 days) in treatment. These factors will be included as time-dependent variables in our primary analysis.

We will also include calendar year as a time-fixed variable to account for potential changes in prescribing practices over time due to the introduction of fentanyl in 2012, emergency declaration in 2016 and relaxation of take-home dosing requirements and flexibility in getting take-home doses due to the pandemic in March 2020.¹⁹ We will also include three time-dependent covariates: use of virtual care, concurrent receipt of short-acting opioids such as hydromorphone tablets alongside methadone^{119 120} and receipt of OAT by delivery. These variables were innovations in treatment delivery to accommodate

clients during the pandemic. Online supplemental table 1 presents the codes that will be used to identify the concurrent conditions.

Sensitivity and subgroup analyses

We will perform sensitivity and subgroup analyses to explore the robustness of the findings and heterogeneity of exposure effects across subgroups. We propose analyses by restricting the cohort and study timeline and reclassifying the outcomes (table 5). Since prescribers' decisions to approve take-home doses based on the assessment of clinical and social stability may not be well reflected in administrative data, we will also conduct sensitivity analyses to deal with residual confounding bias by considering additional time-dependent healthcare variables as well as using the instrumental variable approach^{121 122} (table 5, online supplemental table 2). Our primary focus is to restrict the cohort according to exclusion criteria

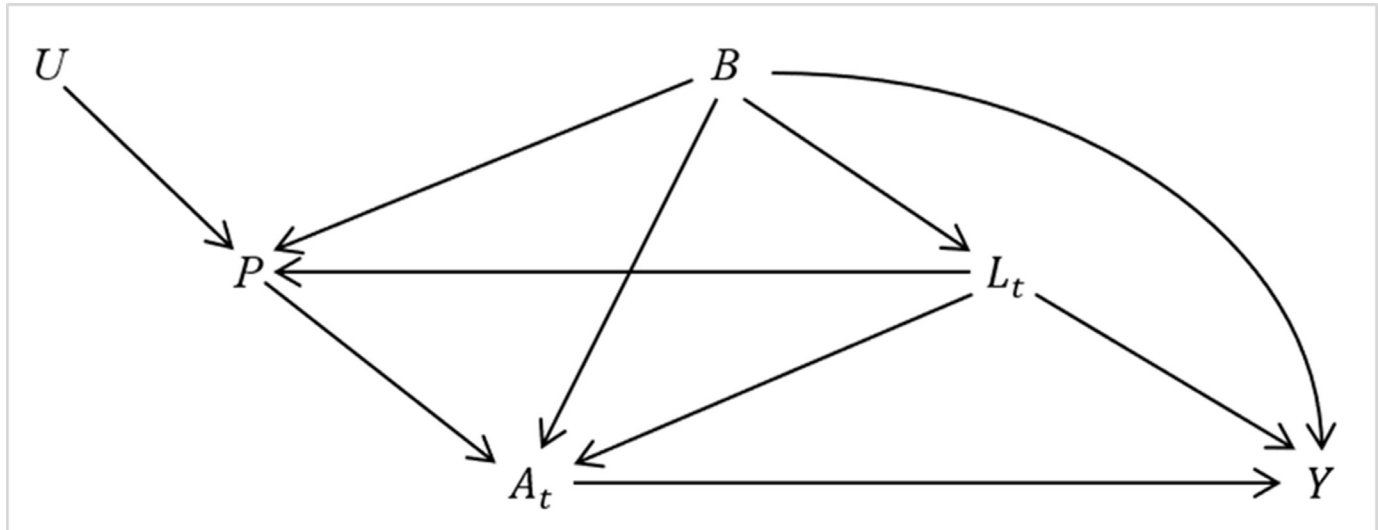


Figure 3 Diagram showing the relationship between the time of initiation of take-home doses (A) and outcome (Y) for individuals who completed induction for methadone in British Columbia, Canada.

Legend: A_t represents exposure strategies (time to initiating take-home doses) at time t , and Y represents the study outcomes (methadone discontinuation, all-cause mortality, acute-care visits) measured after time t . P represents prescribers' decisions to approve methadone take-home doses, and U represents unmeasured covariates which may impact prescribers' decision to approve take-home doses, such as, assessment of clinical and social stability. Baseline/time-fixed confounders B include age at completed induction, sex, place of residence, care from community health centres and calendar year. Time-dependent confounders L_t include unstable housing, receipt of income assistance, Charlson Comorbidity Index, chronic pain, non-opioid substance use disorder, alcohol use disorder, tobacco use disorder, severe mental disorder, sedative medication, prescription opioid other than opioid agonist treatment, psychiatric medication, mental healthcare, drug-related hospitalisation or emergency department visits (except for the acute-care visit outcomes), psychiatric hospitalisations, time to completed induction for methadone, interruptions in methadone doses, methadone dose level, use of virtual care, receipt of treatment by delivery and concurrent receipt of short-acting opioids alongside methadone.

Table 5 Proposed subgroup and sensitivity analyses in exploring the relationship of times for take-home methadone dosing with treatment discontinuation, all-cause mortality and acute-care visits.

Sensitivity and subgroup analyses	Rationale
Sample restriction	
Alternative washout period restrictions	To accommodate the impact of the 30-day washout period in defining prevalent new users, we will consider varying lengths of the washout period, such as a 5-day period (ie, no washout after discontinuation) and a longer 90-day washout period.
Exclude individuals with no fixed address	Individuals with no fixed address indicate housing instability and may not be eligible for take-home doses. We will exclude those individuals with no fixed address at time zero.
Exclude individuals who experience any drug-related ED visit or hospitalisation	Drug-related ED visit or hospitalisation after OAT initiation is an indicator of instability. Individuals who experience any drug-related ED visit or hospitalisation may not be eligible for take-home doses. Our sensitivity analysis will exclude these individuals at time zero.
Exclude individuals with psychiatric hospitalisations	Psychiatric hospitalisation is another indicator of instability. Individuals with psychiatric hospitalisations are less likely to receive take-home doses. Our sensitivity analysis will exclude these individuals at time zero. Pending sufficient data, we might combine 1(c) and 1(d).
Exclude individuals getting care from community health centres	Individuals receiving care from community health centres are often clinically unstable and have multiple comorbidities. Our sensitivity analysis will exclude these individuals getting care from community health centres at time zero.
Exclude those individuals in the rural areas	In some rural regions, pharmacies are closed on weekends. Hence, clinics and pharmacies allow clients to carry their dose for the weekends. We will exclude those individuals in the rural areas where weekend take-home doses may be required.
Include individuals regardless of age	There would be a few individuals aged ≥ 12 and less than 18 years at completed induction and eligible for take-home doses. Including them in the analysis would help us explore the relationship regardless of age and generalise the findings.
Alternative definition of time zero	To accommodate the impact of the 2 weeks of continuous treatment with no dose changes in defining completed induction (time zero), we will consider varying lengths of continuous treatment with no dose changes, such as 3 weeks of continuous treatment with no dose changes.
Alternative definition of time zero as methadone initiation to include those received take-home before completed induction	To assess take-home provision to those early in treatment, we will categorise the exposure as (i) no take-home dosing and take-home dose in (ii) ≤ 4 weeks, (iii) 5–12 weeks, (iv) 13–24 weeks and (v) 25–52 weeks since methadone initiation. The time zero will be set to the date of methadone initiation compared to completed methadone induction for the primary analysis.
Timeline restriction	
Pre- and post-COVID-era data	To provide a comprehensive follow-up history of individuals prior to the disruptions in care and policy changes prompted by the COVID-19 pandemic, we will restrict our follow-up period until 17 March 2020 (the date public health emergency was declared in the province). We will further conduct a sensitivity analysis for the period between 18 March 2020 and 31 December 2022 to address the flexibility in the take-home dosing pattern prompted by the COVID-19 pandemic.
Fentanyl introduction in the illicit drug supply	Fentanyl was introduced in the illicit drug supply in 2014. We will restrict our analysis from 1 January 2014 to explore the robustness of our findings against the introduction of fentanyl in the illegal drug supply.

Continued

Table 5 Continued

Sensitivity and subgroup analyses	Rationale
Outcome classification	
Redefine discontinuation with longer gap times	As seen in other studies and guidelines, ^{92–94 123–125} we will redefine methadone discontinuation with a gap of 14 days compared with 5 days in the primary analysis.
In-treatment mortality	Compared with the primary analysis, where we will consider all-cause mortality over follow-ups, a sensitivity analysis will be conducted with in-treatment mortality. In line with the discontinuation definition and to be consistent with the clinical guideline on dose reversion following discontinuation, ²¹ in-treatment mortality will include deaths occurring up to 5 days after methadone discontinuation.
Other analyses	See online supplemental table 2 for additional sensitivity analyses, such as (a) censoring modelling with a machine learning technique, (b) adding additional healthcare variables with the covariates from the literature, (c) instrumental variable analysis and (d) considering a continuous exposure.
ED, emergency department; OAT, opioid agonist treatment.	

for take-home dosing articulated in the 2017 BC OUD guidelines. Specifically, as guidelines recommend individuals should have a safe place to store take-home doses, we will exclude individuals with no fixed address at time zero. Given the requirement for stability, we will otherwise exclude individuals who experience any drug-related ED visit or hospitalisation within 12 months prior to time zero. We will also exclude those individuals with psychiatric hospitalisations at time zero and residents of rural areas, where weekend take-home doses may be required. Also, we will include all eligible individuals to receive take-home doses regardless of age at completed induction. Though each of these factors may negate the initiation of take-home doses according to clinical guidelines, our primary analysis will include the full sample and employ covariate adjustment for these factors, given that prescribers may not follow guidelines in their prescribing. The 2024 SAMHSA guidelines recommended that individuals could receive take-home doses on the treatment initiation day.⁶⁹ Compared with the primary analysis, where we restricted our analysis to those who completed induction as a measure of stability, we will conduct a sensitivity analysis by including individuals who received take-home doses before the completed induction. All these sensitivity analyses would help us generalise the findings to other settings with different OUD treatment guidelines and healthcare.

We will otherwise restrict the study follow-up period until 17 March 2020. The sensitivity will provide a comprehensive follow-up history of individuals prior to the disruptions in care and policy changes prompted by the COVID-19 pandemic. If sample sizes are sufficient, we will further restrict the study period to 18 March 2020 and 31 December 2022. We will also conduct a sensitivity analysis by redefining methadone discontinuation with a gap of 14 days compared with 5 days in the

primary analysis,^{92–94 123–125} also by fitting the censoring model using a machine-learning method that automates the specification of the weight model¹²⁶ (online supplemental table 2). Any deviations from this protocol will be noted in the final report.

Ethics and dissemination

The linked databases were made available to the research team by British Columbia Ministries of Health and Mental Health and Addiction as part of the response to the provincial opioid overdose public health emergency. The study was classified as a quality improvement initiative. Providence Health Care Research Institute and the Simon Fraser University Office of Research Ethics determined the analysis met criteria for exemption per Article 2.5 of the Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans.¹²⁷

For conducting and reporting research, this study will adhere to international guidelines of the Reporting of Studies Conducted Using Observational Routinely Collected Health Data checklist.¹²⁸ Results will be disseminated to decision-makers, local advocacy groups, and national and international clinical guideline developers. The study findings will also be presented at international conferences and published in peer-reviewed journals. Overall, this study will generate robust evidence regarding the effectiveness of alternative times at which take-home methadone dosing can safely be initiated as seen in real-world clinical practice, in the interest of improving retention in this essential¹²⁹ and life-saving^{7 130} OAT medication.

Patient and public involvement

No patients were directly involved in designing this study. However, the idea for this study was shaped by previous interactions with local advocacy groups

who represent people who use drugs and those who have received OAT. The study was given priority based on feedback obtained from other clinical questions and objectives outlined in the parent grant #R01DA050629. The findings will be shared with local advocacy groups following completion of the analysis.

Author affiliations

- ¹Centre for Advancing Health Outcomes, Vancouver, British Columbia, Canada
- ²School of Population and Public Health, The University of British Columbia, Vancouver, British Columbia, Canada
- ³Faculty of Health Sciences, Simon Fraser University, Burnaby, British Columbia, Canada
- ⁴MRC Biostatistics Unit, University of Cambridge, Cambridge, UK
- ⁵Department of Medicine, The University of British Columbia, Vancouver, British Columbia, Canada
- ⁶British Columbia Centre on Substance Use, Vancouver, British Columbia, Canada
- ⁷Department of Epidemiology, Biostatistics and Occupational Health, McGill University, Montreal, Québec, Canada
- ⁸Department of Statistics, The University of British Columbia, Vancouver, British Columbia, Canada
- ⁹Department of Family Medicine and Emergency Medicine, University of Montreal, Montreal, Québec, Canada
- ¹⁰University of Montreal Hospital Centre, Montreal, Québec, Canada
- ¹¹Department of Statistics and Actuarial Sciences, Simon Fraser University, Burnaby, British Columbia, Canada

X Paxton Bach @paxbach

Acknowledgements We thank members of the Vancouver Area Network of Drug Users, the British Columbia Association of People on Methadone and the Western Aboriginal Harm Reduction Association as well as Dr. Sarah Duffy for their early recommendations, inspiring this line of work. We also thank anonymous reviewers for the excellent feedback on the additional sensitivity analyses.

Contributors BN conceptualised and secured funding for the study. MBH wrote the first draft of the article. MK, JEM, MEK, SS, RWP and PG aided in the methodological development. All authors provided critical revisions to the manuscript and approved the final draft. MBH is responsible for the overall content as the guarantor.

Funding This study was funded by the National Institutes on Drug Abuse grant number R01DA050629.

Disclaimer The funding source was independent of the design of this study and did not have any role during its execution, analyses, interpretation of the data, writing or decision to submit results. The authors had full access to the results in the study and took responsibility for the integrity of the data and accuracy of the analysis.

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, conduct, reporting or dissemination plans of this research.

Patient consent for publication Not applicable.

Provenance and peer review Not commissioned; externally peer-reviewed.

Supplemental material This content has been supplied by the author(s). It has not been vetted by BMJ Publishing Group Limited (BMJ) and may not have been peer-reviewed. Any opinions or recommendations discussed are solely those of the author(s) and are not endorsed by BMJ. BMJ disclaims all liability and responsibility arising from any reliance placed on the content. Where the content includes any translated material, BMJ does not warrant the accuracy and reliability of the translations (including but not limited to local regulations, clinical guidelines, terminology, drug names and drug dosages), and is not responsible for any error and/or omissions arising from translation and adaptation or otherwise.

Open access This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <http://creativecommons.org/licenses/by-nc/4.0/>.

ORCID iDs

Md. Belal Hossain <http://orcid.org/0000-0003-4603-863X>
Shaun Seaman <http://orcid.org/0000-0003-3726-5937>
Paul Gustafson <http://orcid.org/0000-0002-2375-5006>
Bohdan Nosyk <http://orcid.org/0000-0003-2513-3718>

REFERENCES

- 1 Belzak L, Halverson J. The opioid crisis in Canada: a national perspective. *Health Promot Chronic Dis Prev Can* 2018;38:224–33.
- 2 Weedn VW, Elizabeth Zaney M, McCord B, et al. Fentanyl-related substance scheduling as an effective drug control strategy. *J Forensic Sci* 2021;66:1186–200.
- 3 Degenhardt L, Charlson F, Ferrari A, et al. The global burden of disease attributable to alcohol and drug use in 195 countries and territories, 1990–2016: a systematic analysis for the Global Burden of Disease Study 2016. *Lancet Psychiatry* 2018;5:987–1012.
- 4 World Health Organization. Opioid overdose. 2023. Available: <https://www.who.int/news-room/fact-sheets/detail/opioid-overdose> [Accessed 15 Jan 2024].
- 5 Statista. Distribution of deaths from opioid overdose in Canada from 2016 to 2023, by age. 2024. Available: <https://www.statista.com/statistics/1299434/opioid-overdose-deaths-distribution-canada-by-age/#:~:text=In> [Accessed 1 Oct 2024].
- 6 National Center for Health Statistics. U.s.s. overdose deaths decrease in 2023, first time since 2018. 2024. Available: https://www.cdc.gov/nchs/pressroom/nchs_press_releases/2024/20240515.htm#:~:text=The [Accessed 1 Oct 2024].
- 7 Sordo L, Barrio G, Bravo MJ, et al. Mortality risk during and after opioid substitution treatment: systematic review and meta-analysis of cohort studies. *BMJ* 2017;357:j1550.
- 8 Proctor SL, Copeland AL, Kopak AM, et al. A naturalistic comparison of the effectiveness of methadone and two sublingual formulations of buprenorphine on maintenance treatment outcomes: findings from a retrospective multisite study. *Exp Clin Psychopharmacol* 2014;22:424–33.
- 9 Ahmadi J. Methadone versus buprenorphine maintenance for the treatment of heroin-dependent outpatients. *J Subst Abuse Treat* 2003;24:217–20.
- 10 Mattick RP, Breen C, Kimber J, et al. Buprenorphine maintenance versus placebo or methadone maintenance for opioid dependence. *Cochrane Database Syst Rev* 2014;2014:CD002207.
- 11 Johnson RE, Eissenberg T, Stitzer ML, et al. A placebo controlled clinical trial of buprenorphine as a treatment for opioid dependence. *Drug Alcohol Depend* 1995;40:17–25.
- 12 Socías ME, Wood E, Kerr T, et al. Trends in engagement in the cascade of care for opioid use disorder, Vancouver, Canada, 2006–2016. *Drug Alcohol Depend* 2018;189:90–5.
- 13 BC Coroners Service. Unregulated drug deaths- summary. 2024. Available: https://www2.gov.bc.ca/assets/gov/birth-adoption-death-marriage-and-divorce/deaths/coroners-service/statistical/augsept2024_2_unregulated_drug_pdf_of_dashboard.pdf [Accessed 26 Feb 2025].
- 14 Frank D, Mateu-Gelabert P, Perlman DC, et al. “It’s like ‘liquid handcuffs’”: The effects of take-home dosing policies on Methadone Maintenance Treatment (MMT) patients’ lives. *Harm Reduct J* 2021;18:88.
- 15 Adams A, Blawatt S, Magel T, et al. The impact of relaxing restrictions on take-home doses during the COVID-19 pandemic on program effectiveness and client experiences in opioid agonist treatment: a mixed methods systematic review. *Subst Abuse Treat Prev Policy* 2023;18:56.
- 16 Gomes T, Campbell TJ, Kitchen SA, et al. Association Between Increased Dispensing of Opioid Agonist Therapy Take-Home Doses and Opioid Overdose and Treatment Interruption and Discontinuation. *JAMA* 2022;327:846–55.
- 17 Bruneau J, Ahamad K, Goyer M-É, et al. Management of opioid use disorders: a national clinical practice guideline. *CMAJ* 2018;190:E247–57.
- 18 CRISM National Guideline Review Committee. CRISM national guidelines for the clinical management of opioid use disorder. 2018. Available: <https://crism.ca/projects/opioid-guideline/>
- 19 British Columbia Centre on Substance Use, BC Ministry of Health, BC Ministry of Mental Health and Addictions. A guideline for the clinical management of opioid use disorder. 2023. Available: <https://www.bccsu.ca/opioid-use-disorder/>
- 20 SAMHSA Center for Substance Abuse Treatment. TIP 43: medication-assisted treatment for opioid addiction in opioid treatment programs. Available: <https://www.hrsa.gov/behavioral->

- health/tip-43-medication-assisted-treatment-opioid-addiction-opioid-treatment-programs [Accessed 27 Mar 2024].
- 21 Government of British Columbia. A guideline for the clinical management of opioid use disorder. 2017.
 - 22 Government of British Columbia. Methadone maintenance payment program. 2024. Available: <https://www2.gov.bc.ca/gov/content/health/practitioner-professional-resources/pharmacare/pharmacare-publications/pharmacare-policy-manual-2012/pharmacyfees-subsidies-providerpayment/methadone-maintenance-payment-program> [Accessed 15 May 2024].
 - 23 Hongdilokkul N, Krebs E, Zang X, *et al.* The effect of British Columbia's Pharmacare coverage expansion for opioid agonist treatment. *Health Econ* 2021;30:1222–38.
 - 24 Government of British Columbia. Drug coverage for opioid use disorder. 2024. Available: <https://www2.gov.bc.ca/gov/content/health/health-drug-coverage/pharmacare-for-bc-residents/what-we-cover/drug-coverage-opioid-use-disorder> [Accessed 27 Mar 2024].
 - 25 Government of British Columbia. OAT PINs and DINs. 2024. Available: <https://www2.gov.bc.ca/gov/content/health/practitioner-professional-resources/pharmacare/pharmacies/product-identification-numbers/oat-pins-and-dins> [Accessed 27 Mar 2024].
 - 26 Nosyk B, Kurz M, Guerra-Alejos BC, *et al.* Incremental expenditures attributable to daily dispensation and witnessed ingestion for opioid agonist treatment in British Columbia: 2014–20. *Addiction* 2023;118:1376–80.
 - 27 Srivastava A, Kahan M. Methadone induction doses: are our current practices safe? *J Addict Dis* 2006;25:5–13.
 - 28 Liao D-L, Chen P-C, Chen C-H, *et al.* Higher methadone doses are associated with lower mortality in patients of opioid dependence in Taiwan. *J Psychiatr Res* 2013;47:1530–4.
 - 29 Lam V, Latreille S, McLeod A, *et al.* A new framework for methadone carries: a person-centered evidence-informed approach to methadone takehome “carry” dosing. 2023 Available: <https://www.metaphi.ca/wp-content/uploads/CarryPrinciples.pdf>
 - 30 Alberta College of Pharmacy. Opioid agonist therapy (oat) guidelines medication-assisted treatment for opioid use disorder: guidelines for pharmacists and pharmacy technicians. 2022. Available: https://abpharmacy.ca/wp-content/uploads/Guidelines_OAT.pdf
 - 31 Saskatchewan College of Pharmacy Professionals. Opioid agonist therapy (oat) standards. 2020. Available: https://www.saskpharm.ca/document/5871/REF_OAT_Standards.pdf
 - 32 The College of Physicians & Surgeons of Manitoba. Manitoba methadone & buprenorphine maintenance recommended practice. 2024. Available: <https://www.cpsm.mb.ca/prescribing-practices-program/manitoba-buprenorphine-naloxone-recommended-practice-manual>
 - 33 The College of Physicians & Surgeons of Manitoba. Manitoba opioid agonist therapy recommended practice: recommendations for the in-hospital care of individuals with opioid use disorder. 2023. Available: <https://cpsm.mb.ca/prescribing-practices-program/manitoba-buprenorphine-naloxone-recommended-practice-manual>
 - 34 College of Physicians and Surgeons of Ontario. *Methadone Maintenance Treatment Program Standards and Clinical Guidelines* 2011.
 - 35 The College of Physicians and Surgeons of Prince Edward Island. Policy on prescribing methadone maintenance treatment for opioid dependency. 2012. Available: <https://www.cpspei.ca/wp-content/uploads/2012/12/Policy-on-METHADONE-Maintenance-Treatment-for-Opioid-Dependency-Nov-2012.pdf>
 - 36 New Brunswick College of Pharmacists. Methadone maintenance treatment policies and procedures. 2018. Available: <https://nbpharmacists.ca/wp-content/uploads/2022/07/OAT-PracticeDirectiveDocumentationTemplatesJuly2022EN.pdf>
 - 37 College of physicians and surgeons of nova scotia. In: *CPSNS methadone maintenance treatment handbook*. 2012. Available: <https://drugslibrary.wordpress.stir.ac.uk/files/2017/07/guideline-methadone-maintenance.pdf>
 - 38 College of Physicians and Surgeons of Newfoundland and Labrador. Practice guideline: opioid prescribing for opioid use disorder. 2022. Available: <https://cpsnl.ca/wp-content/uploads/Standards-and-Guidelines/Opioid-Prescribing-for-Opioid-Use-Disorder.pdf>
 - 39 Substance abuse and mental health services administration. In: *Medications for Opioid Use Disorder: For Healthcare and Addiction Professionals, Policymakers, Patients, and Families*. 2021. Available: <https://www.ncbi.nlm.nih.gov/books/NBK535268/>
 - 40 Centre for Addiction and Mental Health. Opioid agonist therapy: a synthesis of canadian guidelines for treating opioid use disorder. 2021. Available: <http://www.camh.ca/>
 - 41 Cunningham C, Edlund MJ, Fishman M. The ASAM National Practice Guideline for the Treatment of Opioid Use Disorder: 2020 Focused Update. *J Addict Med* 2020;14:1–91.
 - 42 Kleinman RA, Sanches M. Methadone-involved overdose deaths in the United States before and during the COVID-19 pandemic. *Drug Alcohol Depend* 2023;242:109703.
 - 43 Panwala V, Thorn E, Amiri S, *et al.* Opioid use and COVID-19: a secondary analysis of the impact of relaxation of methadone take-home dosing guidelines on use of illicit opioids. *Am J Drug Alcohol Abuse* 2023;49:597–605.
 - 44 Panwala V, Joudrey P, Kowalski M, *et al.* Changes to methadone maintenance therapy in the United States, Canada, and Australia during the COVID-19 pandemic: A narrative review. *J Subst Use Addict Treat* 2023;152:209086.
 - 45 Kitchen SA, Campbell TJ, Men S, *et al.* Impact of the COVID-19 pandemic on the provision of take-home doses of opioid agonist therapy in Ontario, Canada: A population-based time-series analysis. *Int J Drug Policy* 2022;103:103644.
 - 46 Zlotnick C, Tam T, Robertson MJ. Disaffiliation, substance use, and exiting homelessness. *Subst Use Misuse* 2003;38:577–99.
 - 47 Bouhnik A-D, Chesney M, Carrieri P, *et al.* Nonadherence Among HIV-Infected Injecting Drug Users: The Impact of Social Instability. *JAIDS Journal of Acquired Immune Deficiency Syndromes* 2002;31:S149–53.
 - 48 Akre E-RL, Marthey DJ, Ojukwu C, *et al.* Social Stability and Unmet Health Care Needs in a Community-Based Sample of Women Who Use Drugs. *Health Serv Res Manag Epidemiol* 2021;8:23333928211048640.
 - 49 Substance Abuse and Mental Health Services (SAMHSA). Opioid treatment program (otp) guidance. 2020. Available: <https://www.samhsa.gov/sites/default/files/otp-guidance-20200316.pdf> [Accessed 15 Jan 2024].
 - 50 Amram O, Amiri S, Thorn EL, *et al.* Changes in methadone take-home dosing before and after COVID-19. *J Subst Abuse Treat* 2022;133:108552.
 - 51 Ling W, Stephenson D, Vasti E. Guidelines for physicians working in california opioid treatment programs. 2019. Available: <https://csam-asam.org/wp-content/uploads/csam-guidelines.pdf>
 - 52 Gowing L, Ali R, Dunlop A, *et al.* National guidelines for medication-assisted treatment of opioid dependence. 2014. Available: <https://www.health.gov.au/sites/default/files/national-guidelines-for-medication-assisted-treatment-of-opioid-dependence.pdf>
 - 53 Western Australian Drug and Alcohol Office. Western australian community program for 31 opioid pharmacotherapy (cpop): clinical policies and procedures for the use of methadone and buprenorphine in the treatment of opioid dependence. 2014. Available: <https://www.mhc.wa.gov.au/media/1614/wa-clinical-policies-and-procedures-for-the-use-of-methadone.pdf>
 - 54 New South Wales Ministry of Health. NSW clinical guidelines: treatment of opioid dependence. 2018. Available: <https://www.health.nsw.gov.au/aod/Pages/nsw-clinical-guidelines-opioid.aspx>
 - 55 New Zealand Ministry of Health. New zealand practice guidelines for opioid substitution treatment. 2014. Available: <https://www.health.govt.nz/publication/new-zealand-practice-guidelines-opioid-substitution-treatment-2014>
 - 56 Independent Expert Working Group. Drug misuse and dependence: uk guidelines on clinical management. 2017. Available: https://assets.publishing.service.gov.uk/media/5a821e3340fb62305b92945/clinical_guidelines_2017.pdf
 - 57 National Institute for Health and Care Excellence (NICE). Opioid dependence. 2022. Available: <https://cks.nice.org.uk/topics/opioid-dependence/>
 - 58 Bakouni H, McNulty C, Tatar O, *et al.* Associations of methadone and buprenorphine-naloxone doses with unregulated opioid use, treatment retention, and adverse events in prescription-type opioid use disorders: Exploratory analyses of the OPTIMA study. *Am J Addict* 2023;32:469–78.
 - 59 Jin H, Marshall BDL, Degenhardt L, *et al.* Global opioid agonist treatment: a review of clinical practices by country. *Addiction* 2020;115:2243–54.
 - 60 Heo YA, Scott LJ. Buprenorphine/Naloxone (Zubsolv®): A Review in Opioid Dependence. *CNS Drugs* 2018;32:875–82.
 - 61 Lofwall MR, Walsh SL. A review of buprenorphine diversion and misuse: the current evidence base and experiences from around the world. *J Addict Med* 2014;8:315–26.
 - 62 Saunders EC. Flexible Buprenorphine/Naloxone Treatment Models: Safe and Effective in Reducing Opioid Use Among

- Persons With Prescription Opioid Use Disorder. *Am J Psychiatry* 2022;179:699–701.
- 63 Bickel WK, Amass L, Crean JP, *et al.* Buprenorphine dosing every 1, 2, or 3 days in opioid-dependent patients. *Psychopharmacology (Berl)* 1999;146:111–8.
 - 64 Lam V, Sankey C, Wyman J, *et al.* COVID-19: opioid agonist treatment guidance. 2020. Available: <https://www.camh.ca/-/media/files/covid-19-modifications-to-opioid-agonist-treatment-delivery-pdf.pdf>
 - 65 Lintzeris N, Hayes V, Arunogiri S. Interim guidance for the delivery of medication assisted treatment of opioid dependence in response to covid-19: a national response. 2020. Available: https://www.racp.edu.au/docs/default-source/news-and-events/covid-19/interim-guidance-delivery-of-medication-assisted-treatment-of-opioid-dependence-covid-19.pdf?sfvrsn=e36eeb1a_4
 - 66 Department of Health and Social Care. COVID-19: guidance for commissioners and providers of services for people who use drugs or alcohol. 2020. Available: <https://www.gov.uk/government/publications/covid-19-guidance-for-commissioners-and-providers-of-services-for-people-who-use-drugs-or-alcohol/covid-19-guidance-for-commissioners-and-providers-of-services-for-people-who-use-drugs-or-alcohol>
 - 67 Vecchio S, Ramella R, Drago A, *et al.* COVID19 pandemic and people with opioid use disorder: innovation to reduce risk. *Psychiatry Res* 2020;289:113047.
 - 68 Departament de Salut. Coronavirus sars-cov-2: recomanacions per reduir el risc de contagi de la covid-19 per a centres d'atenció i seguiment de les drogodependències (cas) [fullet]. 2020. Available: <https://scientiasalut.gencat.cat/handle/11351/4799>
 - 69 Substance Abuse and Mental Health Services Administration. Methadone take-home flexibilities extension guidance. 2024. Available: <https://www.samhsa.gov/medications-substance-use-disorders/statutes-regulations-guidelines/methadone-guidance> [Accessed 20 Dec 2024].
 - 70 Brothers S, Viera A, Heimer R. Changes in methadone program practices and fatal methadone overdose rates in Connecticut during COVID-19. *J Subst Abuse Treat* 2021;131:108449.
 - 71 Jones CM, Compton WM, Han B, *et al.* Methadone-Involved Overdose Deaths in the US Before and After Federal Policy Changes Expanding Take-Home Methadone Doses From Opioid Treatment Programs. *JAMA Psychiatry* 2022;79:932–4.
 - 72 McIlveen JW, Hoffman K, Priest KC, *et al.* Reduction in Oregon's Medication Dosing Visits After the SARS-CoV-2 Relaxation of Restrictions on Take-home Medication. *J Addict Med* 2021;15:516–8.
 - 73 Amram O, Amiri S, Panwala V, *et al.* The impact of relaxation of methadone take-home protocols on treatment outcomes in the COVID-19 era. *Am J Drug Alcohol Abuse* 2021;47:722–9.
 - 74 Holland R, Matheson C, Anthony G, *et al.* A pilot randomised controlled trial of brief versus twice weekly versus standard supervised consumption in patients on opiate maintenance treatment. *Drug Alcohol Rev* 2012;31:483–91.
 - 75 Walley AY, Cheng DM, Pierce CE, *et al.* Methadone dose, take home status, and hospital admission among methadone maintenance patients. *J Addict Med* 2012;6:186–90.
 - 76 Wyatt JP, Suen LW, Coe WH, *et al.* Federal and State Regulatory Changes to Methadone Take-Home Doses: Impact of Sociocultural Factors. *Am J Public Health* 2022;112:S143–6.
 - 77 Figgatt MC, Salazar Z, Day E, *et al.* Take-home dosing experiences among persons receiving methadone maintenance treatment during COVID-19. *J Subst Abuse Treat* 2021;123:108276.
 - 78 Madden EF, Christian BT, Lagisetty PA, *et al.* Treatment provider perceptions of take-home methadone regulation before and during COVID-19. *Drug Alcohol Depend* 2021;228:109100.
 - 79 Hunter SB, Dopp AR, Ober AJ, *et al.* Clinician perspectives on methadone service delivery and the use of telemedicine during the COVID-19 pandemic: A qualitative study. *J Subst Abuse Treat* 2021;124:108288.
 - 80 Bardwell G, Small W, Lavalley J, *et al.* "People need them or else they're going to take fentanyl and die": A qualitative study examining the 'problem' of prescription opioid diversion during an overdose epidemic. *Soc Sci Med* 2021;279:113986.
 - 81 Kaufman DE, Kennelley AL, McCall KL, *et al.* Examination of methadone involved overdoses during the COVID-19 pandemic. *Forensic Sci Int* 2023;344:111579.
 - 82 British Columbia Ministry of Health [creator]. Client roster. British Columbia Ministry of health [publisher]. Data Extract. MOH (2024); 2024. Available: <http://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 83 British Columbia Ministry of Health [creator]. PharmaNet. British Columbia Ministry of Health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 84 British Columbia Ministry of Health [creator]. Medical services plan (msp) payment information file. British Columbia Ministry of Health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 85 Canadian Institute of Health Information [creator]. Discharge abstract database (hospital separations). British Columbia Ministry of Health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 86 British Columbia Ministry of Health [creator]. National ambulatory care reporting system (nacrs). British Columbia Ministry of health [publisher]. Data Extract. MOH; 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 87 Ministry of Public Safety and Solicitor General (PSSG) [creator]. BC corrections dataset. british columbia ministry of health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 88 Perinatal Services BC [creator]. British columbia perinatal data registry. years provided: 2000 to 2022. British Columbia Ministry of Health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 89 British Columbia Ministry of Social Development and Poverty Reduction [creator]. Social development and poverty reduction database (sdpr). british columbia ministry of health [publisher]. Data Extract. MOH (2024); 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 90 British Columbia Ministry of Health. Vital statistics deaths. British Columbia Ministry of Health [publisher]. Data Extract. MOH; 2024. Available: <https://www2.gov.bc.ca/gov/content/health/conducting-health-research-evaluation/data-access-health-data-central>
 - 91 Government of British Columbia. Personal health identification. 2023. Available: <https://www2.gov.bc.ca/gov/content/health/health-drug-coverage/msp/bc-residents/personal-health-identification> [Accessed 10 Oct 2024].
 - 92 Piske M, Thomson T, Krebs E, *et al.* Comparative effectiveness of buprenorphine-naloxone versus methadone for treatment of opioid use disorder: a population-based observational study protocol in British Columbia, Canada. *BMJ Open* 2020;10:e036102.
 - 93 Guerra-Alejos BC, Kurz M, Min JE, *et al.* Comparative effectiveness of urine drug screening strategies alongside opioid agonist treatment in British Columbia, Canada: a population-based observational study protocol. *BMJ Open* 2023;13:e068729.
 - 94 Kurz M, Guerra-Alejos BC, Min JE, *et al.* Influence of physician networks on the implementation of pharmaceutical alternatives to a toxic drug supply in British Columbia. *Implement Sci* 2024;19:3.
 - 95 Kurz M, Min JE, Dale LM, *et al.* Assessing the determinants of completing OAT induction and long-term retention: A population-based study in British Columbia, Canada. *J Subst Abuse Treat* 2022;133:108647.
 - 96 Jarvis MA, Wu-Pong S, Kniseley JS, *et al.* Alterations in methadone metabolism during late pregnancy. *J Addict Dis* 1999;18:51–61.
 - 97 Jarvis M, Williams J, Hurford M, *et al.* Appropriate Use of Drug Testing in Clinical Addiction Medicine. *J Addict Med* 2017;11:163–73.
 - 98 Nicholson AB, Watson GR, Derry S, *et al.* Methadone for cancer pain. *Cochrane Database Syst Rev* 2017;2:CD003971.
 - 99 McPherson ML, Walker KA, Davis MP, *et al.* Safe and Appropriate Use of Methadone in Hospice and Palliative Care: Expert Consensus White Paper. *J Pain Symptom Manage* 2019;57:635–45.
 - 100 Zhou Z, Rahme E, Abrahamowicz M, *et al.* Survival Bias Associated with Time-to-Treatment Initiation in Drug Effectiveness Evaluation: A Comparison of Methods. *Am J Epidemiol* 2005;162:1016–23.
 - 101 Farmer R, Mathur R, Bhaskaran K, *et al.* Promises and pitfalls of electronic health record analysis. *Diabetologia* 2018;61:1241–8.
 - 102 Jones M, Fowler R. Immortal time bias in observational studies of time-to-event outcomes. *J Crit Care* 2016;36:195–9.
 - 103 Beyersmann J, Wolkewitz M, Schumacher M. The impact of time-dependent bias in proportional hazards modelling. *Stat Med* 2008;27:6439–54.

- 104 Hernán MA, Sauer BC, Hernández-Díaz S, *et al.* Specifying a target trial prevents immortal time bias and other self-inflicted injuries in observational analyses. *J Clin Epidemiol* 2016;79:70–5.
- 105 Maringe C, Benítez Majano S, Exarchakou A, *et al.* Reflection on modern methods: trial emulation in the presence of immortal-time bias. Assessing the benefit of major surgery for elderly lung cancer patients using observational data. *Int J Epidemiol* 2020;49:1719–29.
- 106 Hernán MA, Robins JM. Using Big Data to Emulate a Target Trial When a Randomized Trial Is Not Available. *Am J Epidemiol* 2016;183:758–64.
- 107 D'Agostino RB, Lee ML, Belanger AJ, *et al.* Relation of pooled logistic regression to time dependent Cox regression analysis: the Framingham Heart Study. *Stat Med* 1990;9:1501–15.
- 108 Robins JM, Hernán MA, Brumback B. Marginal Structural Models and Causal Inference in Epidemiology. *Epidemiology (Sunnyvale)* 2000;11:550–60.
- 109 Karim ME, Hossain MB. Considerations for choosing an imputation method for addressing sparse measurement issues dictated by the study design - An illustration from per-protocol analysis in pragmatic trials. *Research Methods in Medicine & Health Sciences* 2022;3:74–85.
- 110 Mojaverian N, Moodie EEM, Bliu A, *et al.* The Impact of Sparse Follow-up on Marginal Structural Models for Time-to-Event Data. *Am J Epidemiol* 2015;182:1047–55.
- 111 Murray EJ, Hernán MA. Improved adherence adjustment in the Coronary Drug Project. *Trials* 2018;19:158.
- 112 Cole SR, Hernán MA. Constructing inverse probability weights for marginal structural models. *Am J Epidemiol* 2008;168:656–64.
- 113 Cameron AC, Gelbach JB, Miller DL. Robust Inference With Multiway Clustering. *Journal of Business & Economic Statistics* 2011;29:238–49.
- 114 Cole SR, Lau B, Eron JJ, *et al.* Estimation of the standardized risk difference and ratio in a competing risks framework: application to injection drug use and progression to AIDS after initiation of antiretroviral therapy. *Am J Epidemiol* 2015;181:238–45.
- 115 Hernán MA, Robins JM. Estimating causal effects from epidemiological data. *J Epidemiol Community Health* 2006;60:578–86.
- 116 Petersen ML, Porter KE, Gruber S, *et al.* Diagnosing and responding to violations in the positivity assumption. *Stat Methods Med Res* 2012;21:31–54.
- 117 Gran JM, Roysland K, Wolbers M, *et al.* A sequential Cox approach for estimating the causal effect of treatment in the presence of time-dependent confounding applied to data from the Swiss HIV Cohort Study. *Stat Med* 2010;29:2757–68.
- 118 Williams A, Krawczyk N, Hu M-C, *et al.* Retention and critical outcomes among new methadone maintenance patients following extended take-home reforms. *SSRN [Preprint]* 2023.
- 119 Min JE, Guerra-Alejos BC, Yan R, *et al.* Opioid Coprescription Through Risk Mitigation Guidance and Opioid Agonist Treatment Receipt. *JAMA Netw Open* 2024;7:e2411389.
- 120 Provincial Health Services Authority. Unregulated drug poisoning emergency dashboard. 2025. Available: <http://www.bccdc.ca/health-professionals/data-reports/substance-use-harm-reduction-dashboard> [Accessed 9 Jan 2025].
- 121 Robins JM. Analytic methods for estimating HIV-treatment and cofactor effects. *Methodol issues AIDS Behav Res. Published Online* 1993;213–88.
- 122 Shi J, Swanson SA, Kraft P, *et al.* Instrumental variable estimation for a time-varying treatment and a time-to-event outcome via structural nested cumulative failure time models. *BMC Med Res Methodol* 2021;21:258.
- 123 Bell J, Trinh L, Butler B, *et al.* Comparing retention in treatment and mortality in people after initial entry to methadone and buprenorphine treatment. *Addiction* 2009;104:1193–200.
- 124 Morgan JR, Schackman BR, Leff JA, *et al.* Injectable naltrexone, oral naltrexone, and buprenorphine utilization and discontinuation among individuals treated for opioid use disorder in a United States commercially insured population. *J Subst Abuse Treat* 2018;85:90–6.
- 125 Henry-Edwards S, Gowing L, White J, *et al.* Clinical guidelines and procedures for the use of methadone in the maintenance treatment of opioid dependence. 2003.
- 126 Karim ME, Petkau J, Gustafson P, *et al.* On the application of statistical learning approaches to construct inverse probability weights in marginal structural Cox models: Hedging against weight-model misspecification. *Communications in Statistics - Simulation and Computation* 2017;46:7668–97.
- 127 Canadian institutes of health research, natural sciences and engineering research council of canada, social sciences and research humanities council. Tri-council policy statement: ethical conduct for research involving humans. 2018.
- 128 Benchimol EI, Smeeth L, Guttman A, *et al.* The REporting of studies Conducted using Observational Routinely-collected health Data (RECORD) statement. *PLoS Med* 2015;12:e1001885.
- 129 World Health Organization. WHO model lists of essential medicines. 2023. Available: <https://www.who.int/groups/expert-committee-on-selection-and-use-of-essential-medicines/essential-medicines-lists> [Accessed 30 Jan 2024].
- 130 Fugelstad A, Stenbacka M, Leifman A, *et al.* Methadone maintenance treatment: the balance between life-saving treatment and fatal poisonings. *Addiction* 2007;102:406–12.